

SIDEQUEST

The Social Operating System for College Students

Find your people. Build things together. Discover opportunities. Make college count.

A long-term product and engineering vision for a student-first platform combining social discovery, project matchmaking, campus communities, events, career opportunities, and personalized AI.

Project Proposal

Prepared as a flagship software engineering project with the potential to evolve from a college MVP into a real multi-campus product.

1. Executive Summary

SIDEQUEST is a student-first social and collaboration platform designed around the real day-to-day needs of college students. Instead of forcing students to use separate tools for social life, communities, projects, events, networking, and career discovery, SIDEQUEST brings those experiences into one ecosystem.

The central idea is simple: a student's college experience should be represented by more than a resume. SIDEQUEST creates a living student identity based on skills, interests, projects, goals, availability, communities, and collaboration history.

The platform can begin as a focused MVP and gradually evolve into a technically sophisticated system with real-time communication, recommendation engines, semantic search, skill graphs, AI assistance, mobile applications, cloud infrastructure, and multi-college support.

2. The Problem

College students currently distribute their academic, social, project, and career lives across many unrelated platforms.

- WhatsApp and Telegram for class and group communication.
- Instagram and other social platforms for social discovery.
- Discord for communities.
- LinkedIn for professional networking.
- GitHub for projects and collaboration.
- Google Calendar and event pages for activities.
- Notion, Drive and messaging groups for notes and planning.
- Separate platforms for hackathons, internships, coding contests, scholarships and other opportunities.

This fragmentation creates several problems: students do not know who around them shares their interests, finding reliable teammates is difficult, opportunities are scattered, juniors struggle to find useful mentors, and students often have no clear picture of what their peers are building or what they should do next.

3. Product Vision

Vision: Build the digital campus layer where students discover people, form teams, build projects, participate in communities, find opportunities, and receive personalized guidance.

Product principle: SIDEQUEST should feel like a useful student network rather than a college administration portal.

The platform should prioritize discovery, collaboration, community and action. A user should be able to enter with a goal such as “I need a teammate for a hackathon” or “I want to learn Spring Boot with other students” and quickly find relevant people, projects, resources or opportunities.

4. Target Users

- **Students:** discover people, projects, communities, events and opportunities.

- **Project teams:** recruit teammates and coordinate work.
- **Clubs:** build communities and publish events.
- **Mentors and seniors:** help juniors with skills, projects and career preparation.
- **College administrators:** optionally manage verified college spaces and official announcements.

5. Core Product Modules

5.1 Student Identity & Profile

Every student has a living profile rather than a static resume. It can contain skills, interests, projects, goals, availability, communities, achievements, collaboration history and what the student is currently looking for.

Example: a student can mark themselves as a React developer, interested in AI, currently learning Spring Boot, building an app, and looking for a hackathon team.

5.2 People Discovery

Students can search and discover peers by skills, interests, year, college, project interests, availability and goals. SIDEQUEST can eventually rank results using a compatibility engine rather than simple keyword matching.

5.3 Project Matchmaking

A student can create a project idea, describe the required skills, expected time commitment and duration, and recruit teammates. The recommendation system can suggest students based on skills, interests, availability, experience and collaboration compatibility.

5.4 Project Workspace

Each project can contain a team, description, technology stack, tasks, progress, discussions, GitHub repository and recruitment status. This turns SIDEQUEST from a social discovery tool into a place where students actually build together.

5.5 Build in Public

Students can share authentic progress updates about projects, learning and experiments. The tone is intentionally more casual and student-oriented than traditional professional networking. Posts can include milestones, failures, questions, demos and lessons learned.

5.6 Campus Communities

Each college can have structured communities such as departments, years, coding clubs, entrepreneurship clubs, hostel communities, placement groups and interest groups.

5.7 Campus Feed

The feed focuses on useful campus activity: questions, opportunities, project recruitment, events, club announcements, study groups and student progress.

5.8 Events

Students can discover and register interest in hackathons, workshops, club events, seminars, coding contests, conferences and other activities. Events can support reminders and calendar integration.

5.9 Opportunities

SIDEQUEST can aggregate internships, hackathons, scholarships, competitions, research opportunities, fellowships and open-source programs. Instead of only listing them, the platform can eventually estimate a student's match and identify missing skills.

5.10 Mentorship

Students can request mentorship based on specific needs such as DSA, Java, AWS, resumes, projects or placement preparation. Seniors and mentors can define what they are willing to help with.

5.11 Reputation & Collaboration Score

Rather than relying only on follower counts, SIDEQUEST can build a useful collaboration reputation from completed projects, peer endorsements, team contributions, hackathons and reliability signals.

5.12 Anonymous Student Space

A carefully moderated anonymous mode can support candid student conversations. Because anonymity can increase abuse and spam, this module requires reporting, rate limiting, moderation and content-safety mechanisms.

6. AI Layer

AI should be used to make the platform understand student context, not simply as a chatbot wrapper.

6.1 AI Career Mentor

The AI can use a student's skills, projects, goals and target roles to explain what they should learn next, identify gaps and suggest relevant opportunities.

Example: if a student already knows Java, DSA, SQL and REST APIs, the system could identify Spring Boot, JPA and Spring Security as logical next skills for a backend-oriented goal.

6.2 AI College Copilot

The copilot can understand tasks, deadlines, events, projects and learning goals. A student could say: "I have an exam Friday, a hackathon Sunday and an interview Monday," and receive a practical schedule based on available time.

6.3 Semantic Search

Students should be able to search naturally, for example: "Find students who know Python and are interested in computer vision and available this weekend." Embeddings and vector search can support semantic matching.

6.4 Skill-Gap Analysis

The system can compare a student's current skills with a target role, project or opportunity and explain the highest-value gaps.

7. Recommendation & Matching Engine

The matching engine is one of the technically important parts of SIDEQUEST. A simple first version can use weighted scoring; later versions can incorporate embeddings, graph relationships and behavioral signals.

Potential signals include:

- Skill overlap and skill complementarity.
- Shared interests.
- Availability and time commitment.
- Project experience.
- Preferred technologies.
- College, department and year where relevant.
- Previous collaboration quality.
- Project requirements.
- Student goals and target roles.

Example: an AI project requiring frontend, backend, ML and design can be matched with several students whose combined skills cover the team requirements.

8. Skill Graph

SIDEQUEST can represent skills as a connected graph rather than a flat list. Java can connect to OOP, DSA and Spring Boot; Spring Boot can connect to REST APIs, JPA and Spring Security. This allows the platform to recommend learning paths and identify missing skills.

The long-term Student Graph can connect students, skills, projects, opportunities, communities and relationships.

9. Suggested Technical Architecture

The architecture should evolve gradually. The initial implementation should remain simple enough to develop and deploy, while the design should leave room for later services.

Layer	Suggested Technology
Frontend	Next.js / React, TypeScript, Tailwind, component library
Main backend	Java Spring Boot, Spring Security, REST APIs
Supporting services	Node.js where useful for specialized services
Primary database	PostgreSQL
Caching / rate limiting	Redis
Search	PostgreSQL full-text search initially; vector database later
Real-time	WebSockets
AI / ML	LLM APIs, embeddings, Python/ML services where appropriate
Messaging / events	Kafka or a managed queue at a later scale
Storage	AWS S3
Deployment	Docker + AWS
CI/CD	GitHub Actions
Monitoring	CloudWatch and structured application logging

10. High-Level Architecture Flow

Student → Next.js frontend → API gateway/backend → authentication, profile, social, project, event and opportunity modules → PostgreSQL/Redis → recommendation/search/AI services → notifications and real-time communication.

At scale, selected workloads can become independent services connected through events or queues. This should be introduced only when the product actually needs it.

11. Core Data Model

- Users and authentication
- Student profiles
- Skills and skill relationships

- Projects and teams
- Project roles and applications
- Posts, comments and reactions
- Communities and memberships
- Events and registrations
- Opportunities and applications
- Connections and mentorship
- Messages and conversations
- Notifications
- Reports, moderation and audit records

12. Development Roadmap

The project should be built in stages. The goal is not to implement every feature immediately.

Phase 1 — MVP

- Authentication and student onboarding.
- College verification/basic college spaces.
- Student profile with skills, interests and goals.
- People discovery.
- Project creation and team recruitment.
- Basic campus feed.
- Events.
- Basic notifications.

Phase 2 — Collaboration

- Real-time messaging.
- Project workspaces.
- Task and progress tracking.
- Communities and clubs.
- Mentorship.
- GitHub integration.
- Better search and filtering.

Phase 3 — Intelligence

- Recommendation engine.
- Project-team compatibility scoring.
- Opportunity matching.
- Skill-gap analysis.
- Semantic search.
- Skill graph.
- AI career mentor.

Phase 4 — AI & Scale

- AI college copilot.
- Embeddings and vector search.
- Advanced recommendation models.
- Automated moderation assistance.
- Real-time notification infrastructure.
- Event-driven services.

- Load testing and performance optimization.

Phase 5 — Real Product

- Mobile application.
- Multi-college onboarding.
- College and club administration.
- Analytics.
- Production monitoring.
- Public launch and user feedback loops.

13. Security & Trust

Because SIDEQUEST is a social platform, security and trust are core product requirements rather than optional additions.

- JWT/OAuth authentication.
- Role-based access control.
- Secure password hashing.
- Input validation and output sanitization.
- Rate limiting and abuse prevention.
- Account blocking and reporting.
- Content moderation.
- Audit logging for administrative actions.
- Secure secret management.
- Privacy-conscious handling of student data.

14. What Makes SIDEQUEST Different

- It is student-first rather than administration-first.
- The profile represents a student's interests, projects and goals, not only credentials.
- The platform is designed around collaboration and discovery.
- Project matchmaking gives the social layer a practical purpose.
- The AI layer is contextual and connected to the student's activity.
- The recommendation system creates a technical moat beyond CRUD functionality.
- The product can start at one college and grow into a multi-campus network.

15. Success Metrics

If SIDEQUEST is eventually used by real students, success should be measured through actual behavior rather than feature count.

- Number of verified students.

- Weekly and monthly active users.
- Projects created.
- Successful teammate matches.
- Events joined.
- Mentorship connections.
- Opportunities discovered and applied to.
- Projects completed.
- Student retention.
- User-reported usefulness of recommendations.

16. Long-Term Vision

The long-term goal is not to create another student portal. It is to create a digital layer for student life.

A mature SIDEQUEST ecosystem could look like:

Students ↔ People ↔ Skills ↔ Projects ↔ Communities ↔ Events ↔ Opportunities ↔ AI

The network becomes more useful as more students participate. A larger student graph produces better recommendations, more project opportunities, stronger communities and more useful discovery.

17. Why This Is a Strong Flagship Engineering Project

SIDEQUEST can demonstrate substantially more than frontend and backend CRUD development. It provides room to demonstrate system design, Spring Boot, PostgreSQL, Redis, real-time systems, recommendation algorithms, semantic search, AI integration, security, cloud deployment, CI/CD, observability and product thinking.

It also has an important property that many portfolio projects lack: it can be tested with real users. The first target can be a single college. Feedback from actual students can determine which features deserve further development.

18. Suggested Final Resume Positioning

SIDEQUEST — Student Social & Collaboration Platform

A student-first platform for discovering peers, forming project teams, participating in campus communities, finding opportunities and receiving personalized AI guidance. The platform can combine Spring Boot, PostgreSQL, Redis, real-time communication, recommendation services, semantic search, AI and AWS into a scalable architecture.

19. Final Product Statement

SIDEQUEST is a social operating system for college students where students can find their people, form teams, build projects, discover campus life and career opportunities, and receive personalized guidance based on who they are and what they want to achieve.

The project should begin small, prove that students want it, and then grow technically and geographically. The ambition is to build something that is not merely impressive in a portfolio, but useful enough that real students choose to keep using it.